

Rina Bao

Publication List

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Book Chapter

1. Bao, R., He, S., Grant, E., and Ou, Y., "Linking Deep Learning and MRI to Brain Disorders," in *Handbook of the Biology and Pathology of Mental Disorders*, Springer.

Journal Publications

1. **[Scientific Data'25]** Bao, R., Song, Y., Bates, S. V., Weiss, R. J., Foster, A. N., Jaimes, C., Sotardi, S., Zhang, Y., Hirschtick, R. L., Grant, P. E., and Ou, Y., "BOSton Neonatal Brain Injury Data for Hypoxic Ischemic Encephalopathy (BONBID-HIE): I. MRI and Lesion Labeling," *Scientific Data*, vol. 12, Article 53, 2025. doi. IF: 8.7.
2. **[IEEE TMI'26]** Bao, R., Foster, A. N., Song, Y., Vyas, R., Kesri, A., Toubal, I. E., Soltani Kazemi, E., Rahmon, G., Kucukpinar, T., Almansour, M., Ho, M.-L., Palaniappan, K., Ninalga, D., Koirala, C. P., Mohapatra, S., Schlaug, G., Wodzinski, M., Muller, H., Ellis, D. G., Aizenberg, M. R., Aydın, M. A., Abidinli, E., Unal, G., Tahmasebi, N., Punithakumar, K., Song, T., Peng, Y., Bates, S. V., Hirschtick, R., Grant, P. E., and Ou, Y., "BONBID-HIE 2023: Lesion Segmentation Challenge in BOSton Neonatal Brain Injury Data for Hypoxic Ischemic Encephalopathy," *IEEE Transactions on Medical Imaging*, vol. 45, no. 4, pp. 1711–1725, 2026. doi. IF: 9.8.
3. **[MIA'25]** Bao, R., Weiss, R. J., Bates, S. V., Song, Y., He, S., Li, J., Bjornerud, A., Hirschtick, R. L., Grant, P. E., and Ou, Y., "PARADISE: Personalized and regional adaptation for HIE disease identification and segmentation," *Medical Image Analysis*, vol. 102, Article 103419, 2025. doi. IF: 11.8.
4. **[IEEE JSTSP'24]** Bao, R., Zhao, Y., Srivastava, A., Frazier, S., and Palaniappan, K., "Lightweight SDE-Net Fusing Model-Based and Learned Features for Computational Histopathology," *IEEE Journal of Selected Topics in Signal Processing*, vol. 18, no. 6, pp. 1123–1137, Sept. 2024. doi. IF: 13.7.
5. **[IEEE TGRS'26]** Bao, R., Palaniappan, K., and Zhao, Y., "PointGrad: A Unified Spatially Varying Tensor Gradient Operator for Point Cloud Segmentation," *IEEE Transactions on Geoscience and Remote Sensing*, 2026. doi. IF: 8.6.
6. **[IEEE Sensors Journal'24]** Bao, R., Palaniappan, K., Zhao, Y., and Seetharaman, G., "GLSNet++: Global and Local-Stream Feature Fusion for LiDAR Point Cloud Semantic Segmentation Using GNN Demixing Block," *IEEE Sensors Journal*, vol. 24, no. 7, pp. 11610–11624, 2024. doi. IF: 4.7.
7. **[npj Computational Materials'21]** Hajilounezhad, T.*, Bao, R.*, Palaniappan, K., Bunyak, F., Calyam, P., and Maschmann, M. R., "Predicting carbon nanotube forest attributes and mechanical properties using simulated images and deep learning," *npj Computational Materials*, vol. 7, Article 134, 2021. doi. *Equal contribution. IF: 13.0.
8. **[Nature Methods'23]** Maška, M., Ulman, V., Delgado-Rodriguez, P., Gómez-de-Mariscal, E., Nečasová, T., Guerrero Peña, F. A., Ing Ren, T., Meyerowitz, E. M., Scherr, T., Löffler, K., Mikut, R., Guo, T., Wang, Y., Allebach, J. P., Bao, R., Al-Shakarji, N. M., Rahmon, G., Toubal, I. E., Palaniappan, K., Lux, F., Matula, P., Sugawara, K., Magnusson, K. E. G., Aho, L., Cohen, A. R., Arbelle, A., Ben-Haim, T., Riklin Raviv, T., Isensee, F., Jäger, P. F., Maier-Hein, K. H., Zhu, Y., Ederra, C., Urbiola, A., Meijering, E., Cunha, A., Muñoz-Barrutia, A., Kozubek, M., and Ortiz-de-Solórzano, C., "The Cell Tracking Challenge: 10 years of objective benchmarking," *Nature Methods*, vol. 20, pp. 1010–1020, 2023. doi. IF: 51.7.
9. **[JAMIA Open'25]** Hsiao, C.-H., Foster, A. N., McDonald, S. A., Vyas, R., Ashraf, A., Bao, R., Tran, L., Kesri, A., Darzidehkalani, E., Soldatelli, M. D., Auman, J. O., Soul, J. S., Chalak, L. F., Cotten, C. M., Shankaran, S., Laptook, A. R., Grant, P. E., and Ou, Y., "Data Harmonization Framework for Neonatal Hypoxic-Ischemic Encephalopathy Studies," *JAMIA Open*, vol. 8, no. 5, Article ooaf086, 2025. doi. IF: 3.2.

Conference Publications

1. **[ICML'25]** Bao, R., Dong, S., Chen, Z., He, S., Grant, E., and Ou, Y., "Visual and Domain Knowledge for Professional-level Graph-of-Thought Medical Reasoning," *Proceedings of the 42nd International Conference on Ma-*

chine Learning, 2025. **Spotlight, Top 2.6%**.

2. **[MICCAI'25]** Rao, A., Shukla, A., Jia, B., Ou, Y., and Bao, R., "Spatial Prior-Guided Boundary and Region-Aware 2D Lesion Segmentation in Neonatal Hypoxic Ischemic Encephalopathy," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, LNCS, Springer, 2025. [doi](#).
3. **[AIPR'25]** Hatuwal, B. K., Bao, R., Ho, M.-L., and Palaniappan, K., "HE3D-Net: Hypoxic Ischemic Encephalopathy Diagnosis and Lesion Segmentation Using 3D Heterogeneous Ensemble Models," in *Applied Imagery Pattern Recognition Workshop*, LNCS 16446, pp. 495–507, Springer, 2026. [doi](#).
4. **[ICCV W'21]** Bao, R., Al-Shakarji, N. M., Bunyak, F., and Palaniappan, K., "DMNet: Dual-stream marker guided deep network for dense cell segmentation and lineage tracking," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, pp. 3361–3370, 2021. **Best Paper Award**.
5. **[ISBI'26]** Rao, A., Shukla, A., Jia, B., Siddharth, S., and Bao, R., "Obscuration to Clarity: Bone Suppression for Enhanced Localization in Pneumothorax Segmentation of Chest Radiographs," in *IEEE International Symposium on Biomedical Imaging*, 2026. **Oral**. Preprint: [doi](#).
6. **[ISBI'25]** Bao, R., He, S., Grant, E., and Ou, Y., "AGE2HIE: Transfer Learning from Brain Age to Predicting Neurocognitive Outcome for Infant Brain Injury," in *IEEE International Symposium on Biomedical Imaging*, 2025.
7. **[ISBI'25]** He, S., Bao, R., Grant, P. E., and Ou, Y., "Relation U-Net," in *IEEE International Symposium on Biomedical Imaging*, pp. 1–4, 2025. [doi](#).
8. **[ISBI'25]** Li, J., He, S., Bao, R., Bjørnerud, A., Grant, E., and Ou, Y., "Using Off-Target-Disease Data to Improve Target-Disease Lesion Segmentation," in *IEEE International Symposium on Biomedical Imaging*, pp. 1–4, 2025.
9. **[AIPR'19]** Bao, R., Palaniappan, K., Zhao, Y., Seetharaman, G., and Zeng, W., "GLSNet: Global and local streams network for 3D point cloud classification," *IEEE Applied Imagery Pattern Recognition Workshop*, 2019. **Oral**.

Conference Abstracts

1. **[PAS'26]** Bao, R., Hsiao, C.-H., Bates, S. V., Weiss, R., Gollub, R., Grant, E., and Ou, Y., "Anatomy-Normalized ZADC Maps Improves Detection of Neonatal Brain Injury in HIE," Pediatric Academic Societies Meeting, 2026. **Oral**.
2. **[PAS'26]** Bao, R., Grant, E., and Ou, Y., "Agentic Artificial Intelligence for Hypoxic-Ischemic Encephalopathy Prognosis through Clinically Guided Reasoning on Neonatal MRI," Pediatric Academic Societies Meeting, 2026. **Oral**.
3. **[SPR'23]** Bao, R., Grant, E., and Ou, Y., "2-Year neurocognitive outcome prediction of Hypoxic Ischemic Encephalopathy in neonates with brain MRIs," The Society for Pediatric Radiology Annual Meeting, 2023.
4. **[Nutrition'23]** Bao, R., Wu, D., Vyas, R., Kuchan, M., Lasekan, J., Sutton, B., Grant, E., Morton, S., and Ou, Y., "Maternal Dietary Nutrition Interact With Demographic and Socioeconomic Information To Impact Offspring Neurocognition by Early School Age," *Nutrition*, 2023.
5. **[PAS'23]** Bao, R., Grant, E., and Ou, Y., "Clinical Data Predicting Neonatal and 2-year Outcome Following Hypoxic Ischemic Encephalopathy," Pediatric Academic Societies Meeting, 2023.
6. **[PAS'25]** Ou, Y., Kesri, A., Bao, R., Hsiao, C.-H., Vyas, R., Hussain, M. A., McDonald, S., Auman, J., Foster, A., Darzi, E., Soldatelli, M., Shankaran, S., Laptook, A., Cotten, M., and Grant, E., "Combining 47 clinical and neuroimaging variables to predict adverse 18–22-month outcomes for Hypoxic Ischemic Encephalopathy in Neonatal Research Network trials," Pediatric Academic Societies Meeting, 2025.
7. **[ASPNR'25]** Kesri, A., Bao, R., Hsiao, C.-H., Vyas, R., Hussain, M. A., McDonald, S., Auman, J., Shankaran, S., Laptook, A., Cotten, M., Grant, E., and Ou, Y., "The role of expert MRI scores in predicting adverse 18–22-month outcomes for Hypoxic Ischemic Encephalopathy in Neonatal Research Network trials," Annual Meeting of American Society of Pediatric Neuroradiology, 2025.

Manuscripts Under Review

1. **[Science Advances]** Bao, R., Zhao, Y., Ou, Y., and Palaniappan, K., "Brain-Inspired Visual Primitives Enable Cross-Domain Learning from Few Examples," major revision at *Science Advances*.
2. **[npj Science of Food]** Bao, R., Wu, D., Vyas, R., Kuchan, M. J., Lasekan, J. B., Leyshon, B., Gopinath, A., Sutton, B. P., Grant, P. E., Morton, S. U., and Ou, Y., "Maternal nutrition and sociodemographic factors determinants jointly impact neurocognition of breastfed offspring at early school age: A pilot learning study," under review at *npj Science of Food*.
3. **[Epilepsia]** Kesri, A. *, Bao, R. *, Valery, C., Iannotti, I., McLaren, J., Pinto, A. L., Comi, A. M., Ou, Y., "Machine

Learning Could Pre-Symptomatically Predict Seizure Onset by 2 Years for Sturge-Weber Syndrome", under review at *Epilepsia*. (* equal contribution)

4. **[AI in Radiology]** Bao, R., Foster, A. N., Song, Y., Vyas, R., Xu, R., Piao, Z., Choi, J., Gu, Y., Ninalga, D., Choi, J. W., Hatuwal, B. K., Kazemi, E. S., Palaniappan, K., Rao, A., Shukla, A., Bhargava, J., Dhouib, H., Pan, Z., Liang, X., Yildirim, A. S., Şahin, Y. H., Baek, S., Morton, S. U., Bates, S. V., Hirschtick, R., Grant, P. E., and Ou, Y., "BONBID-HIE 2024: Lesion Segmentation and Outcome Prediction from Boston Neonatal Brain Injury MRI in Hypoxic-Ischemic Encephalopathy – Review and Results," under review at *AI in Radiology*.
5. **[MIA]** Darzi, E., Hsiao, C.-H., Bao, R., Weiss, R. J., Bates, S. V., Song, Y., He, S., Li, J., Bjornerud, A., Hirschtick, R. L., Grant, P. E., and Ou, Y., "SparseMed3D: Foundation Models for Sparse Instance Medical Segmentation," under review at *Medical Image Analysis*.

Preprints

1. Bao, R. and Ou, Y., "BOston Neonatal Brain Injury Dataset for Hypoxic Ischemic Encephalopathy (BONBID-HIE): II. 2-year Neurocognitive Outcome and NICU Outcome," arXiv:2411.03456, 2024.
2. Bao, R., Darzi, E., He, S., Hsiao, C.-H., Hussain, M. A., Li, J., Bjornerud, A., Grant, E., and Ou, Y., "Foundation AI Model for Medical Image Segmentation," arXiv:2411.02745, 2024. [doi](#).
3. He, S., Bao, R., Grant, P. E., and Ou, Y., "U-Netmer: U-Net meets Transformer for medical image segmentation," arXiv:2304.01401, 2023.
4. He, S., Bao, R., Li, J., Stout, J., Bjornerud, A., Grant, P. E., and Ou, Y., "Computer-vision benchmark segment-anything model (SAM) in medical images: Accuracy in 12 datasets," arXiv:2304.09324, 2023.